Individual Pathways

Activity 10.1

Transforming and transferring energy doc-6093

1. Energy is defined as the ability to do work. Energy can be described as either potential or kinetic. Complete the following statements about these forms of energy.

   (a) All objects that are moving have …… energy.

   (b) All substances and objects possess …… energy. This is also called stored energy.

   There are four types of potential energy: chemical, elastic, gravitational and nuclear. Identify the types of potential energy in the examples below.

   (c) A battery has …… energy.

   (d) The sun generates its energy by …… energy.

   (e) When an elastic band is stretched, it has …… potential energy.

   (f) A pen placed on the end of a bench has …… potential energy.

   There are five types of kinetic energy: motion, light, sound, heat and electrical. Identify the types of kinetic energy in the examples below.

   (g) The energy of runner in the 100 m sprint at the Olympics .....

   (h) The energy produced when a trumpet is played .....

   (i) The main energy given off by the blue flame of a Bunsen burner .....

   (j) The energy used to power a toaster .....

   (k) The main energy given off by a torch .....

2. Heat can be transferred by conduction, convection and radiation. In the example below, the sentences have been separated. Rejoin the sentences and state which type of heat movement the sentence refers to.

		Conduction, convection or radiation?
When particles very close together, such as in a solid, vibrate …	… the heat is moved from particle to particle.	
Heat causes the particles to gain energy …	… and spread apart, as in a liquid.	
Heat can be transferred without the …	… need for particles, such as the heat from the sun.	

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3. Using a diagram, describe at least four ways in which a person could make their house more energy efficient.

4. In the following matrix, use a tick to indicate whether the objects are luminous or non-luminous. Some objects may be both luminous and non-luminous; for example, when a torch is turned on, it is luminous, but when it is turned off, it is non-luminous.

Object	**Luminous**	**Non-luminous**
Sun		
Diamond		
Moon		
Mars		
Light globe		
Yellow Bunsen burner flame		
TV screen		
Cat's eye		

5. How would you make the sound of a guitar string:
   (a) deeper?
   (b) have a higher pitch?
   (c) vibrate faster?

6. A music teacher complained that the music room he taught in had too much reverberation.
   (a) What is reverberation?
   (b) Explain how the music teacher could overcome this problem.

7. 'In space, no-one can hear you scream.'
   (a) Is this famous line accurate? Explain your answer.
   (b) If the quote was 'In space, no-one can see you scream,' would it still be accurate? Explain your answer.

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